Total Questions: 35

LEVEL – II

Time: 1.5 hr.
Total Marks: 70

SECTION - 1	• This section contains **FIFTEEN (15)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer. • For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.
SECTION - 2	• This section contains **EIGHT (08)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s). • For each correct answer you will get **TWO** marks, if all the options are correct. If the correct answers chosen are less than the actual correct answers, partial marking (basis on weightage) will be given. • There are **NO NEGATIVE** marks for wrong answers.
SECTION - 3	• This section contains **NINE (09)** questions of PASSAGE TYPE OR COMPREHENSION TYPE. • Under each paragraph there will be three questions that has **FOUR** options. **ONLY ONE** of these four options is the correct answer. • For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for
SECTION - 4	• This section contains **THREE (03)** questions. These questions are **MATCHING TYPE BASED QUESTIONS.** • For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers

Section - 1

1. **The perimeter of a quadrilateral is 50m. If the lengths of three of its sides are 12m, 15m, and 10m, what is the length of the fourth side, x?**

   (A) 11 m (B) 12 m

   (C) 13 m (D) 14 m

2. **Two mobile network towers transmit signals along paths represented by the equations: $4x + 2y = 8$; $2x + y = 4$**

   Based on their graphical representation on a coordinate plane, what can be concluded about these two signal paths?

   (A) Parallel lines

   (B) Intersecting lines

   (C) Coincident lines

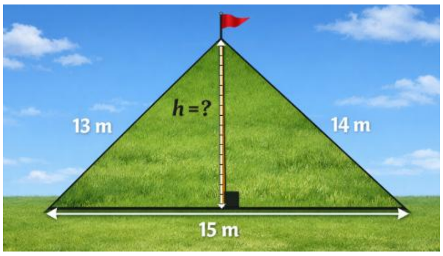
   (D) Perpendicular lines

3. A point lies 5 units left of origin and 3 units above x-axis. Which quadrant does it lie in?

(A) I (B) II

(C) III (D) IV

4. A triangular field has sides 13 m, 14 m, 15 m. A pole is to be placed perpendicular to the longest side. Find the height corresponding to side 15 m.



(A) 10.4 m (B) 11.2 m

(C) 12 m (D) 9.6 m

5. The polynomial

$p(x) = x^4 - 2x^3 + 3x^2 - ax + b$ when divided by $(x-1)$ and $(x+1)$ leaves remainders 5 and 19 respectively. Find the remainder when $p(x)$ is divided by $(x-2)$

(A) 10 (B) 12

(C) 15 (D) 20

6. A carpenter stretches a string tightly between two nails fixed on a wooden board.

He assumes the shortest path between these two nails is the stretched string. Which Euclidean postulate justifies this assumption?

(A) A line can be extended indefinitely

(B) A straight line can be drawn joining any two points

(C) All right angles are equal

(D) A circle can be drawn with any centre and radius

7. A bank account shows a balance of ₹ –3500. To bring the balance to zero, what transaction must be done?

(A) Deposit ₹3500

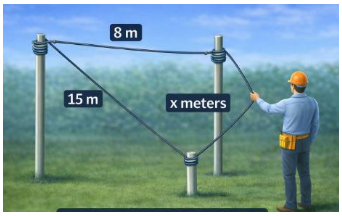
(B) Withdraw ₹3500

(C) Deposit ₹7000

(D) No change required

8. A cable technician is fixing a triangular network between three poles. Two cables measure 8 m and 15 m, while the third cable has length x meters. To ensure proper signal stability, the cables must form a closed triangular connection.

Find all possible values of x.



(A) x = 6 (B) x = 7

(C) x = 22 (D) x = 24

9. A drone camera rotates from its initial position and stops after turning 135° to capture a wide-angle image. The engineer notes that the rotation is more than a right angle but less than a straight line.

Which type of angle is formed?

(A) Acute angle (B) Right angle

(C) Obtuse angle (D) Reflex angle

10. A water tank filling model is given by $p(x) = x^3 - 4x^2 + 5x - 2$ .Find the remainder when it is divided by $x - 2$.

(A) 0 (B) 2 (C) -2 (D) 4

11. In a cyclic quadrilateral ABCD if AB is the diameter of the circle and $\angle ADC = 130^\circ$ then the measure of $\angle BAC$ is:

(A) 40° (B) 50°

(C) 90° (D) 130°

12. **Two adjacent angles are formed when a solar panel is adjusted. One angle measures $(3x - 10)°$ and the other measures $(2x + 20)°$. Find the value of x.**

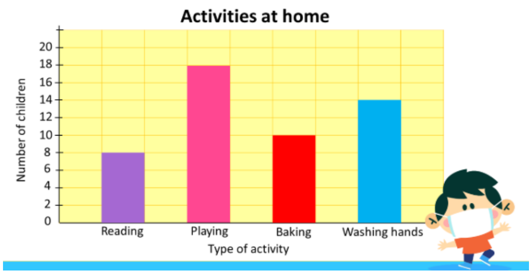
(A) 10 (B) 16 (C) 20 (D) 34

13. **A point M is 4 units away from the origin and lies on the negative y-axis. Another point N has coordinates (3, 0). The shortest distance between M and N is:**

(A) 7 units (B) 1 unit

(C) 5 units (D) $\sqrt{7}$ units

14. **Observe the graph carefully.**



If 6 students move from Washing hands to Baking, the values change.
Compare the new values of Baking and Playing.

(A) Baking > Playing

(B) Baking = Playing

(C) Baking < Playing

(D) Cannot determine

15. **A student claims that $\sqrt{9} + \sqrt{2}$ is rational. Which statement is correct?**

(A) Always rational

(B) Always irrational

(C) Depends on value

(D) Cannot be determined

Section - 2

16. **A chocolate bar is divided into two unequal pieces. One piece is clearly larger than the other. Which statements are always true?**

(A) Whole chocolate is greater than each part

(B) A part can be greater than the whole

(C) Whole equals sum of parts

(D) Smaller part equals whole

17. **A classroom floor is designed such that all angles are 90°. Opposite sides are equal and diagonals are drawn. Engineers check symmetry and congruency.**

Which are correct?

(A) Diagonals are equal

(B) Diagonals bisect each other

(C) Diagonals are perpendicular

(D) Each diagonal forms congruent triangles

18. **Consider the polynomial**

$f(x) = ax^2 + bx + c$. If $f(1) = f(2) = 0$.

then which of the following are true?

(A) $b = -3a$

(B) $c = 2a$

(C) $a + b + c = 0$

(D) $f(x) = a(x^2 - 3x + 2)$

19. **In a triangle, one interior angle is 50°. The exterior angle at that vertex is considered. Identify correct statements.**

(A) Exterior angle = 130°

(B) Sum of opposite interior angles = 130°

(C) Exterior angle = sum of all interior angles

(D) Exterior angle equals opposite angles

20. **Three roads connect three villages forming a triangle of sides 7 km, 24 km, and 25 km. Engineers calculate the enclosed land area. They also study the type of triangle formed.**

Which are correct?

(A) Triangle is right-angled

(B) Area = 84 km^2

(C) Semi-perimeter = 28 km

(D) Area can be found without height

21. At a construction site, two rods intersect forming four angles. One angle is marked $(4x - 10)°$ and its vertically opposite angle is $(3x + 20)°$.

Which statements are correct?

(A) $x = 30$

(B) Each of those angles is 110°

(C) Adjacent angles are supplementary

(D) All four angles are equal

22. An ice-cream cone has radius 3 cm and height 4 cm. It is completely filled with ice-cream.

Which of the following are correct?

(A) Volume of cone = 12π cm^3

(B) Curved surface area = 15π cm^2

(C) If radius is doubled, volume becomes 8 times

(D) Slant height = 5 cm

23. The mean of 5 observations is 20. Four observations are 10, 15, 25, 30.

Which are correct?

(A) Fifth observation = 20

(B) Sum of observations = 100

(C) Mean increases if fifth observation increases

(D) Median = 20

Section - 3

Passage 1:

A chemist mixes two solutions: Solution A costs ₹100 per litre and Solution B costs ₹250 per litre.

He prepares 10 litres of mixture worth ₹1750.

Let x and y be litres of A and B respectively.

The mixture must satisfy both quantity and cost conditions.

The chemist wants exact proportions.

24. What is the correct pair of equations?

(A) $x + y = 10$ and $100x + 250y = 1750$

(B) $x - y = 10$ and $100x + 250y = 1750$

(C) $x + y = 1750$ and $100x + 250y = 10$

(D) $100x + 250y = 10$ and $x - y = 1750$

25. The quantity of Solution B used is:

(A) 3 litres (B) 4 litres

(C) 5 litres (D) 6 litres

26. The system of equations represents:

(A) Parallel lines

(B) Intersecting lines

(C) Coincident lines

(D) No solution

PASSAGE-2

A tent is designed as a triangular prism.

The triangular base has sides 7 cm, 24 cm, and 25 cm. The length of the tent is 30 cm.

The company calculates canvas required and air capacity inside the tent.

27. Area of triangular base is:

(A) 70 cm^2 (B) 84 cm^2

(C) 96 cm^2 (D) 120 cm^2

28. Volume of the tent is:

(A) 2100 cm^3 (B) 2520 cm^3

(C) 3000 cm^3 (D) 3600 cm^3

29. Total surface area is:

(A) 1680 cm^2 (B) 1900 cm^2

(C) 1848 cm^2 (D) 2000 cm^2

PASSAGE-3

A city planner maps a rectangular park on a coordinate grid. The four corners are plotted as A(2, 3), B(8, 3), C(8, 7), and D(2, 7). A fountain is placed exactly at the midpoint of diagonal AC. A walking path is drawn joining the fountain to vertex B.

The planner wants to check distances and symmetry before construction.

30. The coordinates of the fountain are:

(A) (5, 5) (B) (4, 6)

(C) (6, 4) (D) (3, 5)

31. The length of diagonal AC is:

(A) 5 units (B) 6 units

(C) 10 units (D) $\sqrt{52}$ units

32. The distance between the fountain and point B is:

(A) $\sqrt{13}$ units (B) $\sqrt{18}$ units

(C) $\sqrt{10}$ units (D) $\sqrt{8}$ units

Section - 4

33.

Column I (Situation)	Column II (Result)
(a) A rope measures $\sqrt{144}$meters	(i) 0.625
(b) Decimal expansion of $\frac{5}{8}$	(ii) 6
(c) Sum of $\sqrt{2} + (-\sqrt{2})$	(iii) 12
(d) Product of $\sqrt{3} \times \sqrt{12}$	(iv) 0

(A) a–ii, b–i, c–iv, d–iii

(B) a–ii, b–iv, c–iii, d–i

(C) a–i, b–ii, c–iii, d–iv

(D) a–iii, b–i, c–iv, d–ii

34. A robot travels between points. Compare distances.

Column I (Pair of Points)	Column II (Distance Type)
(a) A(1, 4) to B(4, 6)	(i) $\sqrt{13}$ units
(b) C(–2, –4) to D(1, 2)	(ii) $\sqrt{25}$ units
(c) E(0, 0) to F(6, 8)	(iii) 10 units
(d) G(2, –1) to H(5, 3)	(iv) $\sqrt{45}$ units

(A) a–ii, b–i, c–iii, d–iv

(B) a–ii, b–iv, c–iii, d–i

(C) a–i, b–iv, c–iii, d–ii

(D) a–ii, b–i, c–iv, d–iii

35.

Column I (Situations)	Column II (Values)
(a) A cuboid has length 5 cm, breadth 4 cm, height 3 cm. Find its volume	(i) 100 cm^3
(b) A prism has base area 20 cm^2 and height 5 cm. Find its volume.	(ii) 216 cm^2
(c) A cube of side 6 cm. Find its total surface area.	(iii) 90 cm^3
(d) A triangular prism has base area 15 cm^2 and height 6 cm. Find its volume.	(iv) 60 cm^3

(A) a–iv, b–i, c–ii, d–iii

(B) a–ii, b–iv, c–iii, d–i

(C) a–i, b–ii, c–iii, d–iv

(D) a–ii, b–i, c–iv, d–iii